

VisualThink-VLA: Efficient Visual Thinking for Vision-Language-Action Policies

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Abstract

Pretrained vision-language-action (VLA) policies typically operate on raw images and language alone, even when inexpensive structured perception cues are readily available. Existing attempts to enrich intermediate reasoning often either inject dense auxiliary visual evidence, which can slow closed-loop control, or rely on verbose textual chain-of-thought, which adds autoregressive overhead and is not always the most natural medium for vision-first reasoning. This work introduces **VISUALTHINK-VLA**, a sparse soft-evidence interface for frozen VLA policies that supports *think with images* through a **compact structured evidence layer** retained after empirical feature screening: static bbox/edge cues, dynamic motion cues, and instruction-grounded relation cues, while excluding heavier or low-utility channels such as depth and segmentation from the final candidate set. The method is backbone-agnostic at the evidence and routing level and dynamically activates only a sparse subset of evidence channels at each step, using *blended masks* and *distillation* from a dense FULLSOFT teacher. In addition, the paper constructs **EvidenceTrace-VLA**, a derived evidence-trace dataset built on top of open-source robot trajectories, and introduces **counterfactual faithfulness diagnostics** for testing whether selected evidence, generated rationales, and action quality remain aligned. Evaluation spans a broad multi-dataset control suite covering real-world robot trajectories, simulation suites, long-horizon tasks, and an evidence-trace audit layer, with comparisons against publicly available reasoning-augmented VLA baselines. The results indicate that selective structured evidence offers a competitive and efficient alternative to newer reasoning-oriented VLA pipelines while making visual evidence usage substantially easier to supervise and audit across diverse datasets.

1 Introduction

Recent vision-language-action (VLA) policies show that large multimodal models can be adapted to robot control by mapping observations and instructions directly to discretized action tokens (Kim et al., 2024; Brohan et al., 2022, 2023; Mees et al., 2024). Yet real robot observations often contain compact structure that is easy to extract but hard for a generic VLA to use reliably from raw pixels alone: **coarse object localization, geometric edges, short-horizon motion cues, and instruction-grounded spatial relations**. A natural response is to expose more intermediate visual signals to the policy, or to ask the model to externalize longer textual reasoning traces. For real robot control, however, neither direction is fully satisfactory. Dense auxiliary evidence can expand per-step computation and slow the closed-loop policy, while verbose chain-of-thought adds autoregressive overhead and need not be the most natural medium for vision-first reasoning.

This concern is reinforced by recent work on **visual planning**, which argues that language is not always the best substrate for visually grounded reasoning and shows that purely image-based intermediate planning can outperform text-space planning on spatial tasks (Xu et al., 2025). These observations point to a more targeted question for VLA control: rather than adding more text or more dense side information indiscriminately, **which compact visual cues should be surfaced, and how should they be used efficiently across tasks and manipulation stages?** At present, there is still limited systematic guidance on which image-grounded features are actually worth consulting, and on how their usage should vary across broad manipulation phases such as approach, contact, and placement.

The present study examines this problem in the setting of a frozen VLA policy that must consume

compact, image-grounded intermediate evidence in a way that is both *effective* and *selective*. Two requirements matter simultaneously. **First**, the interface must be strong enough for structured evidence to improve the action model itself. **Second**, the policy should avoid consulting every available cue at every step, because dense evidence usage is unnecessary for many decisions and obscures the evidence actually used by the model.

This problem is framed as enabling a VLA policy to *think with images*. Concretely, the policy consults compact image-grounded intermediate evidence before action decoding, such as prioritizing an edge map over motion cues or activating a relation cue for a language-conditioned manipulation step. The framing keeps the reasoning medium visual rather than forcing it through a long textual trace, while making the intermediate computation explicit and measurable by exposing which visual evidence is used and when.

The starting point is that **prompt-text evidence injection is a weak interface for VLA policies whose action decoder is trained on ordinary image-language prompts**. In preliminary experiments with OPENVLA, concatenating structured evidence into prompts degraded action prediction and entangled two distinct design questions: *which evidence is useful* and *how that evidence should be injected*. The proposed interface therefore treats evidence not as extra language, but as a small set of **learned soft evidence states** derived from compact structured evidence vectors and inserted before action decoding. For token-based VLA backbones, the implementation used here inserts these states as soft tokens in the multimodal sequence; in the experiments, OPENVLA serves as the concrete instantiation. This yields a dense reference model denoted as **FULLSOFT**.

Dense evidence, however, is not always necessary. Some tasks benefit from motion-heavy cues, others from simple edges or relation features, and some can be solved by a single strong cue. A broader pool of candidate channels is therefore screened first, and the final retained set is $\mathcal{C} = \{\text{bbox}, \text{edge}, \text{motion}, \text{relation}\}$, covering object location, boundary shape, temporal change, and language-conditioned spatial layout while discarding heavier cues such as depth and segmentation that were less consistently selected or less cost-effective in the experiments. This leads to the main method, **VISUALTHINK-VLA**, which adds **sparse evidence routing** on top of the soft in-

terface. A learned gate predicts which channels to activate, and the sparse adapter is trained using *blended masks* and *distillation* from the FULLSOFT teacher. The resulting system keeps the same frozen VLA backbone but becomes **selective** about which side-channel evidence it consumes. OPENVLA is used as the primary experimental backbone because it is an open token-based VLA with a real action-decoding path, but the evidence bank, router, trace supervision, and quality-governance layer are not specific to OPENVLA.

The paper is organized around a **broad multi-dataset control suite**, the **EvidenceTrace-VLA** supervision-and-diagnosis layer, and a focused real-robot closed-loop study rather than a single benchmark. The main control suite spans real-world datasets, simulator suites, and long-horizon tasks, including **BridgeData V2**, **Fractal**, **RoboTurk**, the **LIBERO** subsets, and **UT Austin MUTE**; EvidenceTrace-VLA then anchors the route, rationale, and faithfulness analyses. Across these settings, the evidence suggests that a single sparse-evidence recipe can improve a real VLA policy without materially sacrificing efficiency, while a channel-grounded trace dataset paired with counterfactual diagnostics makes evidence usage easier to supervise and audit.

Contributions. The main contributions are threefold:

1. **A compact think-with-images evidence layer.** The final channel set $\mathcal{C} = \{\text{bbox}, \text{edge}, \text{motion}, \text{relation}\}$ is distilled through empirical feature screening: it retains object and edge cues, adds dynamic motion and instruction-grounded relation, and removes heavier or weakly used channels such as depth and segmentation from the routed set.
2. **A sparse soft-evidence routing recipe.** VISUALTHINK-VLA dynamically selects which evidence channels to consult at each step and stabilizes this selection with *blended masks* and *teacher distillation*.
3. **An evidence-trace dataset and faithfulness diagnostics.** EvidenceTrace-VLA is constructed as a derived evidence-trace dataset built from open-source robot trajectories and paired with **counterfactual faithfulness diagnostics**, so that routed evidence usage can

be supervised, audited, and interpreted rather than treated as an opaque latent decision.

Method details appear in Section 3, and the full empirical study appears in Sections 4 to 6.

2 Related Work

Vision-language-action policies and reasoning-augmented VLAs. Large-scale robot policies increasingly rely on language-conditioned sequence models trained on heterogeneous demonstration corpora. RT-1 and RT-2 established the scaling pattern for image-language control (Brohan et al., 2022, 2023), while Open X-Embodiment, Octo, and OPENVLA pushed open-source generalist robot policies trained on broad mixtures of robot data (Open X-Embodiment Collaboration, 2023; Mees et al., 2024; Kim et al., 2024). More recent VLA work augments action prediction with explicit intermediate reasoning or stronger grounding, including ECoT, Fast ECoT, DeepThinkVLA, and InternVLA-M1 (Zawalski et al., 2024; Duan et al., 2025; Yin et al., 2025; Chen et al., 2025). These methods form the most relevant external baselines because they ask the policy to reason before acting, but they do so through textual reasoning traces, thought reuse, hybrid reasoning-action decoding, or spatially guided grounding. The focus here is narrower and more systems-oriented: can a frozen VLA policy recover some of the benefits of explicit reasoning through sparse, image-grounded evidence routing, without paying the full runtime cost of generating long reasoning traces? The implementation instantiates this question with OPENVLA, but the evidence and routing design is stated at the level of VLA backbones more generally.

Structured visual evidence for robot control. Object-centric and geometry-aware representations have long been useful for robot learning and visuomotor control. VIOLA, for example, shows that object proposals and structured object-level reasoning can strengthen manipulation policies (Zhu et al., 2022). At the same time, recent visual-planning work suggests that language is not always the best substrate for vision-first reasoning: Xu et al. (2025) show that purely image-based intermediate planning can outperform text-space planning on spatial tasks. The present work is motivated by both lines of research, but asks a narrower question: which compact visual cues

are actually worth exposing to a VLA policy, and how should they be injected without turning the control loop into a dense or text-heavy reasoning pipeline? The evidence channels are built from standard perception back-ends such as OWL-ViT, MiDaS-family depth, SAM2, and Qwen2.5-VL (Minderer et al., 2022; Ranftl et al., 2022; Birkel et al., 2023; Ravi et al., 2024; Lin et al., 2025); these components are not themselves the novelty, but they provide a realistic pool from which compact routed evidence can be selected.

Conditional computation and sparse routing. Sparse activation and conditional computation have been studied extensively in mixture-of-experts and related architectures (Shazeer et al., 2017). The routing problem considered here is smaller in scale but similar in spirit: the model should activate only the evidence channels that matter for the current decision. Unlike standard MoE work, however, the router does not select among parameter-heavy experts inside the language model. Instead, it selects among externally computed evidence channels and feeds them through a lightweight adapter into an otherwise fixed VLA backbone; the implementation here realizes this adapter as soft tokens for OPENVLA.

3 Method

The method section compares **three ways of using structured evidence** with a frozen VLA policy and turns the comparison into a concrete operational notion of *think with images*. The framework is formulated for a generic pretrained VLA backbone; OPENVLA is the concrete backbone used in the main implementation and matched re-evaluation.

Evidence channels. For each observation, the pipeline precomputes **four compact evidence channels**. **bbox** encodes coarse object localization from OWL-ViT detections, and **edge** captures local shape and boundary structure from SAM2 masks. In addition to these relatively static cues, **motion** captures short-horizon temporal change through frame-difference statistics, which is useful for contact, displacement, and phase transitions, while **relation** encodes instruction-grounded spatial geometry between the target object, nearby objects, and goal-side anchors, which is useful for deciding *where* the robot should move an object rather than only *what* object exists. Im-

portantly, this four-channel set is the outcome of a broader empirical screening process. Earlier experiments also considered additional cues such as monocular depth, segmentation-style dense perception, and heavier visual descriptors, but some of them were rarely selected by the learned router, produced less stable gains on the real policy path, or introduced extra extraction cost that was not justified by downstream improvement. The final design therefore retains bbox, edge, motion, and relation as a compact evidence set that is expressive enough to cover object localization, boundary geometry, temporal change, and instruction-conditioned spatial layout, while remaining small enough to keep routing and evidence injection efficient. Each channel is compressed into a **small fixed-length vector** rather than passed as an image-sized tensor. This keeps the side-channel lightweight and makes channel-wise routing explicit.

Think-with-image interpretation. Under the interpretation adopted here, *think with images* means that action prediction is conditioned not only on the raw image encoder state but also on a small set of **explicitly image-grounded intermediate cues**. These cues are visual in origin but easier to route, analyze, and sparsify than full image features. In this view, **FULLSOFT** corresponds to always consulting every available intermediate cue, while **VISUALTHINK-VLA** consults only the cues judged necessary for the current decision. For example, in a language-conditioned pick-and-place step, a relation cue or coarse box may be most useful for grounding the target before contact, a local edge or motion cue may be more diagnostic during contact, and only one of these cues may remain necessary after grasp. The objective is therefore not to verbalize a chain of reasoning, but to make the intermediate visual evidence **explicit and controllable**.

3.1 A soft-evidence interface for frozen VLA policies

The first key design choice is to **avoid prompt-text evidence injection**. Instead, the pretrained VLA backbone is frozen and a lightweight adapter is trained to map structured evidence vectors into a small set of **learned soft evidence states**. The generic requirement is that these states condition the policy before action decoding while preserving the original observation, instruction, and action

head. In token-based VLA models this can be implemented as soft tokens inserted between visual embeddings and task-prompt tokens; in the OPEN-VLA instantiation used here, the soft evidence tokens are inserted into the multimodal sequence before action-token decoding. This design directly augments the real policy computation while keeping the original action decoding path intact. It also provides a concrete implementation of image-grounded intermediate computation inside a VLA model, where evidence states act as *explicit visual side information* that can be either always active or selectively routed.

Figure 1 summarizes the resulting pipeline. At inference time, the observation and instruction are encoded by the base VLA, the evidence bank is routed by the learned gate, and the selected evidence is mapped into soft evidence states that condition the frozen policy before action decoding. In the OPENVLA implementation these states are soft tokens injected into the frozen multimodal sequence. The dashed supervision arrows indicate **training-only** route and trace signals from the second-stage interpretable and verifiable supervision setup; they do not introduce additional inference-time modules.

Let $\mathcal{C} = \{\text{bbox}, \text{edge}, \text{motion}, \text{relation}\}$ denote the final candidate set of structured evidence channels retained after this empirical screening. This notation is useful because both dense evidence injection and sparse routing can be expressed over the same channel universe: **FULLSOFT** uses all $c \in \mathcal{C}$, whereas **VISUALTHINK-VLA** predicts a subset of $\mathcal{C}$ for each step and only keeps the selected channels active. In other words, $\mathcal{C}$ is deliberately designed to be **broad enough** to include complementary static, dynamic, and language-conditioned geometric cues, yet **small enough** that the router is not forced to choose among a long tail of weak or expensive channels. For each channel $c \in \mathcal{C}$, the method produces a vector x_c and projects it into the language-model hidden space. The adapter then outputs a small bank of global soft evidence tokens plus one token per channel. **The backbone remains frozen throughout training; only the adapter parameters are learned.**

3.2 Three compared systems

BASEVLA-only. This baseline runs the frozen pretrained VLA backbone with the **original image and instruction inputs** and **no external evidence**.

Table 1: Compact comparison of the three systems studied in this work.

System	Uses evidence	Routing	Distill
BASEVLA-only	No	None	No
FULLSOFT	All channels	Dense	No
VISUALTHINK-VLA	Selected channels	Sparse learned gate	Yes

In the experiments, this row is instantiated with OPENVLA.

FULLSOFT. FULLSOFT enables **all four evidence channels for every sample**. It measures the value of the soft interface itself when evidence is always available, i.e., when the model always thinks with the full image-grounded evidence set.

VISUALTHINK-VLA. VISUALTHINK-VLA uses a **learned gate** to activate only a **sparse subset of channels**. The gate is trained separately from the VLA adapter using low-cost context and structured evidence supervision, then reused to generate dynamic masks for the soft interface. This separation keeps the real policy path simple while still obtaining a learned routing policy. In this framing, this is *selective image-grounded thinking*: the policy does not verbalize a chain of thought, but it does decide which visual side-evidence is worth consulting.

3.3 Unified sparse routing recipe

The unified recipe used throughout the paper has **three ingredients**.

(1) Unified sequence-text gate. The same final unified gate is used across datasets. At a high level, it consumes low-cost visual context, instruction/query text features, and stage-conditioned routing priors, then predicts channel-selection scores. The output is a **sparse hard mask** $m^{\text{hard}} \in \{0, 1\}^{|C|}$ together with corresponding **soft probabilities** $m^{\text{soft}} \in [0, 1]^{|C|}$.

(2) Blended soft masks. Pure hard routing was too brittle in practice, especially on difficult datasets such as `taco_play`. During dynamic adapter training, the method therefore uses a **blended mask**

$$\tilde{m} = (1 - \alpha)m^{\text{hard}} + \alpha m^{\text{soft}}, \quad (1)$$

with $\alpha = 0.35$. Intuitively, this value is large enough to leak a small amount of probability mass from suppressed channels, which stabilizes optimization, but small enough that the routing behavior remains clearly sparse rather than degenerating

into a dense average. Empirically, lower values made the dynamic adapter too brittle, while higher values blurred the distinction between FULLSOFT and VISUALTHINK-VLA.

(3) Distillation from FULLSOFT. VISUALTHINK-VLA is further trained with logits distillation from the pretrained FULLSOFT adapter. Let p_f^τ and p_d^τ denote the teacher and student action-token distributions at temperature τ . The dynamic objective is

$$\mathcal{L}_{\text{dyn}} = \mathcal{L}_{\text{action}} + \lambda_{\text{distill}} \tau^2 \text{KL}(p_f^\tau \| p_d^\tau), \quad (2)$$

with $\lambda_{\text{distill}} = 0.2$ and $\tau = 1.5$. The role of these parameters is straightforward: $\tau > 1$ softens the teacher distribution so that the student sees more than only the top action token, while a moderate λ_{distill} keeps distillation strong enough to stabilize sparse routing without overwhelming the task loss. Distillation is important empirically because it closes much of the gap between a sparse dynamic adapter and its dense FULLSOFT teacher.

3.4 Evidence-Trace Supervision and Faithfulness Diagnostics

To make the routed visual evidence easier to inspect, the pipeline derives an **Evidence-Trace** record for each sample from the existing feature manifest, dynamic channel masks, and counterfactual channel utilities. Each trace stores the instruction, action label, selected channels, a compact text description of each channel’s evidence, a short visual rationale, an action-intent summary, a utility ranking over channels, and the binary route mask. Crucially, the rationale is **channel-grounded**: it names the selected evidence types and attaches short template-based descriptions to the corresponding measured signals, rather than generating unconstrained free-form CoT.

The method also adds an optional auxiliary supervision head to the soft-evidence adapter. Given the pooled evidence state and the current mask, stage, and step-ratio features, this head predicts a multi-label channel target derived from the Evidence-Trace route mask. When enabled, the

total dynamic loss becomes

$$\mathcal{L}_{\text{dyn+trace}} = \mathcal{L}_{\text{dyn}} + \lambda_{\text{trace}} \mathcal{L}_{\text{BCE}}(\hat{r}, r), \quad (3)$$

where $r \in \{0, 1\}^{|C|}$ is the Evidence-Trace channel target and $\hat{r}$ is the auxiliary prediction. This head is used only as a training regularizer; the main action decoding path is unchanged.

Finally, three counterfactual faithfulness diagnostics are defined. *Top-channel removal* zeros out the highest-ranked active evidence channel and measures the resulting success drop. *Channel shuffling* swaps selected channel features, especially relation and motion, across samples. *Route-rationale agreement* compares the channels named in the generated rationale with the active route mask and the top utility-ranked channels. Together, these diagnostics test whether *think with images* is merely descriptive or whether the stated evidence is tied to the model’s actual routed inputs.

Trace taxonomy and quality governance. Because EvidenceTrace-VLA is derived automatically from heterogeneous robot datasets, the pipeline adds a deterministic governance layer before using traces for supervision or audit. Each trace is assigned a compact taxonomy along four primary axes: manipulation stage, task primitive, evidence-dependence pattern, and difficulty level. The stage label is inherited from the routing dataset; the primitive label is obtained from rule-based instruction parsing; the evidence-dependence label is computed from the route mask and counterfactual utility distribution; and the difficulty score combines trajectory length, stage diversity, relation-word complexity, instruction complexity, utility entropy, and route sparsity. The same pass also checks structural validity, channel-key consistency, route/selected-channel agreement, utility-rank consistency, rationale-route mention recall, and basic action/instruction quality. The output decision is one of `keep_release`, `keep_weighted`, `relabel`, `keep_internal`, or `drop`. This governance layer is used only for dataset construction, sample weighting, and benchmark selection; it does not add any inference-time module.

4 Experimental Setup

Benchmark roles and empirical hierarchy. The empirical study is organized around four complementary evaluation tracks rather than

a single undifferentiated benchmark suite. **First**, it reports a **broad multi-dataset control benchmark table** spanning real-world datasets (BridgeData V2, Fractal, RoboTurk), simulator suites (LIBERO-Object, LIBERO-Goal, LIBERO-Spatial), and long-horizon tasks (LIBERO-Long, UT Austin MUX). This serves as the primary benchmark summary in the main paper. **Second**, the evaluation uses **EvidenceTrace-VLA** as a *supervision-and-audit layer*: it does not replace the control benchmarks, but instead tests whether routed evidence, written rationales, and counterfactual interventions remain aligned. **Third**, the paper includes a focused *real-robot closed-loop evaluation* to assess whether the efficiency motivation survives actual deployment-time control. **Fourth**, additional RLDS/OXE datasets used during development—such as `viola`, `stanford_hydra`, and `taco_play`—are retained as appendix-level transfer evidence beyond the datasets already shown in the main benchmark table.

Baselines and comparison policy. The main external comparisons target recent **reasoning-augmented VLA baselines with public code or checkpoints**: **ECoT** (Zawalski et al., 2024), **Fast ECoT** (Duan et al., 2025), **DeepThinkVLA** (Yin et al., 2025), and **InternVLA-M1** (Chen et al., 2025). For the method family studied here, the results report a frozen BASEVLA baseline, a dense FULLSOFT reference, and sparse VISUALTHINK-VLA. BASEVLA is instantiated by OPENVLA, which provides an open token-based VLA with a real action-decoding path; the proposed evidence bank, router, trace construction, and governance procedure are stated at the VLA level. When public checkpoints and evaluation code are available, the evaluation uses matched re-evaluation on the overlapping benchmarks and measures average step latency together with task success on the local hardware; otherwise the affected entries are marked as paper-reported or unavailable. This reporting policy matters because the baselines differ in backbone, reasoning medium, benchmark coverage, and public reproducibility.

Code release. The project code, configuration files, and reproducibility scripts will be released in an anonymized supplementary archive during review and in a public project repository after acceptance.

Table 2: EvidenceTrace-VLA governance axes. The taxonomy is used to construct cleaner supervision and audit subsets, while preserving a reproducible account of why each sample is retained, downweighted, reviewed, or removed.

Axis	Labels	Signal source
Stage	approach / grasp / place / other	Existing route dataset stage labels
Primitive	pick / place / push / open / close / rearrange / inspect / other	Rule-based instruction parsing
Evidence dependence	bbox / edge / motion / relation dominant; balanced; weak	Route mask and counterfactual utility
Difficulty	L1–L5	Horizon, stage count, instruction relations, utility entropy, sparsity

Structured evidence and EvidenceTrace-VLA.

The evidence channels are extracted offline. **bbox** uses OWL-ViT (Minderer et al., 2022); **edge** uses SAM2 (Ravi et al., 2024); **motion** uses frame differences; and **relation** is a lightweight instruction-grounded geometry feature. Heavier candidates such as monocular depth, segmentation-style dense perception, and text-heavy schema variants are evaluated during channel screening, but the final routed model uses only the empirically retained four-channel set. To make routed evidence trainable and auditable, the paper constructs **EvidenceTrace-VLA**, a derived step-level annotation layer over existing robot trajectories. Each trace stores the route mask, selected evidence names, counterfactual utility ranking, compact channel snippets, a short channel-grounded visual rationale, and an action-intent summary. Figure 2 shows the full construction path from raw frames and instruction, through evidence extraction and dynamic routing, to the final released record schema. The route mask supplies direct supervision for sparse selection, while the utility ranking and rationale support the faithfulness diagnostics used later in the paper.

Trace taxonomy and quality-control protocol.

For the EvidenceTrace-VLA supervision-and-audit track, not all automatically generated traces are treated as equally reliable. After trace construction, a deterministic governance pass assigns taxonomy labels and quality decisions to every row. The taxonomy slices samples by stage, task primitive, dominant evidence pattern, route sparsity, temporal span, instruction complexity, and difficulty level. The quality pass checks required fields, channel-key consistency, route/selected-channel agreement, utility-rank consistency, rationale-route mention recall, and basic action/instruction validity. The pipeline

then exports three nested subsets: **Full-Clean** for broad training and distributional statistics, **HQ-Trace** for trace-supervised refinement, and **Gold-Faithfulness** for high-confidence audit experiments. Counts and source-wise retention rates are reported after all source manifests are finalized.

Real-robot closed-loop protocol. The real-robot study follows the small-suite deployment style commonly used in open VLA and robot-learning evaluations (Brohan et al., 2022, 2023; Kim et al., 2024; Zawalski et al., 2024), but is tailored to the question tested here: whether sparse routed visual evidence remains practical when the policy is executed in a live control loop. The study uses a tabletop manipulation setup with one external RGB camera and the robot wrist camera available to the policy stack. Each episode begins from a randomized but task-feasible scene, receives a natural-language instruction, and runs until success, timeout, or a safety stop. The task set is intentionally small but diagnostic: object-centric pick-and-place, contact-rich insertion or closing, relation-sensitive placement, distractor-heavy object selection, and a multi-step long-horizon task. Each task family compares the same three policy paths used in the offline study—BASEVLA, FULLSOFT, and VISUALTHINK-VLA—without changing the backbone checkpoint. Unlike the offline proxy, this track reports both inference-side latency and real wall-clock completion time, because evidence extraction and routing can affect deployment-time control cadence.

Metrics by evaluation track. The primary control metric is `success_rate`. For efficiency, the main benchmark table reports `avg_step_latency_s`, the average inference-side latency per control step over non-failed samples. This metric includes prompt/image preparation, optional CoT generation, evidence extraction

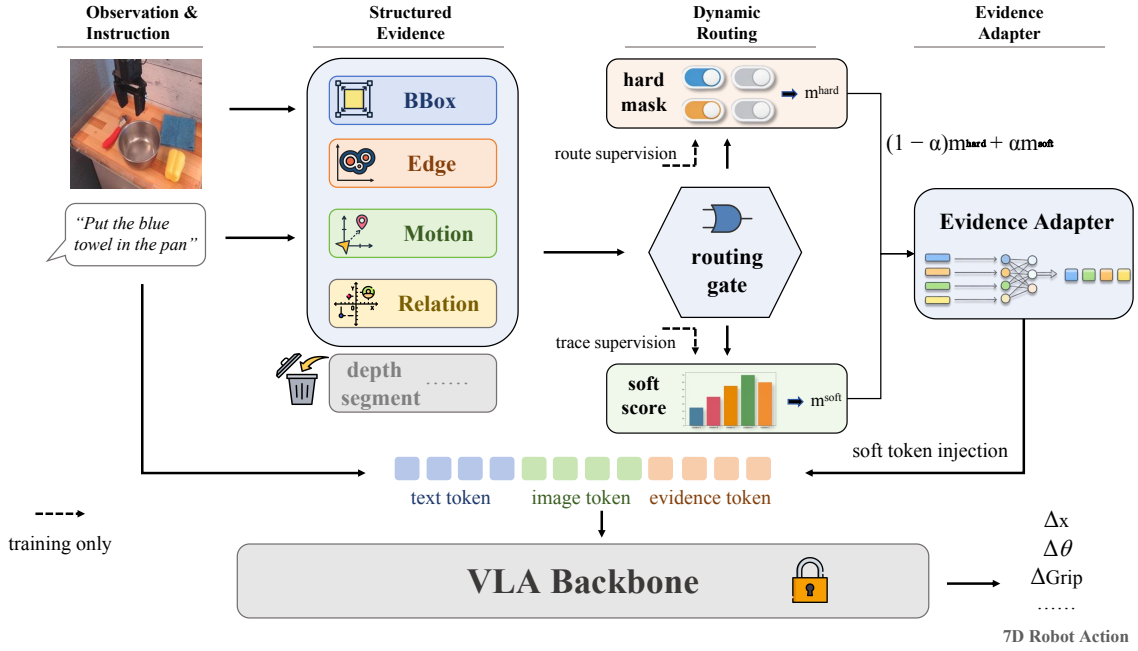


Figure 1: Overview of the VISUALTHINK-VLA pipeline. The observation and instruction are encoded by a frozen VLA backbone, while an external evidence bank provides four structured channels: bbox, edge, motion, and relation. A learned routing gate predicts a sparse hard mask together with soft scores; the effective evidence mask is then mapped by the evidence adapter into soft evidence states that condition the policy before action decoding. In the OPENVLA instantiation used here these states are injected as soft tokens between image tokens and prompt tokens. The dashed arrows denote **training-only** route and trace supervision used in the second-stage interpretable and verifiable training setup; the main inference path remains the solid black path.

when it is executed online, routing, policy forward passes, and action decoding, but excludes environment reset and simulator stepping. It is used as the primary runtime column in the grouped benchmark table because it is directly comparable across datasets while avoiding distortion from failed or timed-out episodes. The evaluation also retains `avg_completion_time_s` and `timeout_penalized_completion_time_s` as supplementary episode-level efficiency diagnostics, especially for the real-robot closed-loop study. On the main benchmark suite, the supporting analyses additionally report the average number of selected channels and per-channel keep rates. For EvidenceTrace-VLA, the primary score is a trace-benchmark metric aggregated from routed-evidence alignment and faithfulness signals; the analysis then unpacks its components with **top-channel removal drop**, **relation/motion shuffle drop**, **route-rationale agreement**, and **utility mention rate**. This separation of metrics reflects the two-part empirical goal of the paper: better control on the one hand, and auditable evidence usage on the other.

Stage-conditioned routing analysis. Because one of the central claims is that different evidence types matter at different points in the trajectory, the paper reports stage-conditioned routing statistics over approach, grasp, and place. To avoid duplicating the main control table, the main text presents this analysis through the governed EvidenceTrace-VLA aggregate in Figure 5, while expanded benchmark-specific routing tables are left to the appendix.

Default VISUALTHINK-VLA recipe. Unless otherwise stated, VISUALTHINK-VLA uses the same configuration across the primary benchmarks: the final unified sequence-text gate, blended masks with $\alpha = 0.35$, logits distillation from the FULLSOFT teacher, and the second-stage route/trace supervision described in Section 3.4. The ablation section then isolates these ingredients one by one: channel screening justifies the final evidence bank, recipe ablations justify the optimization choices, and interpretability ablations justify the role of route and trace supervision.

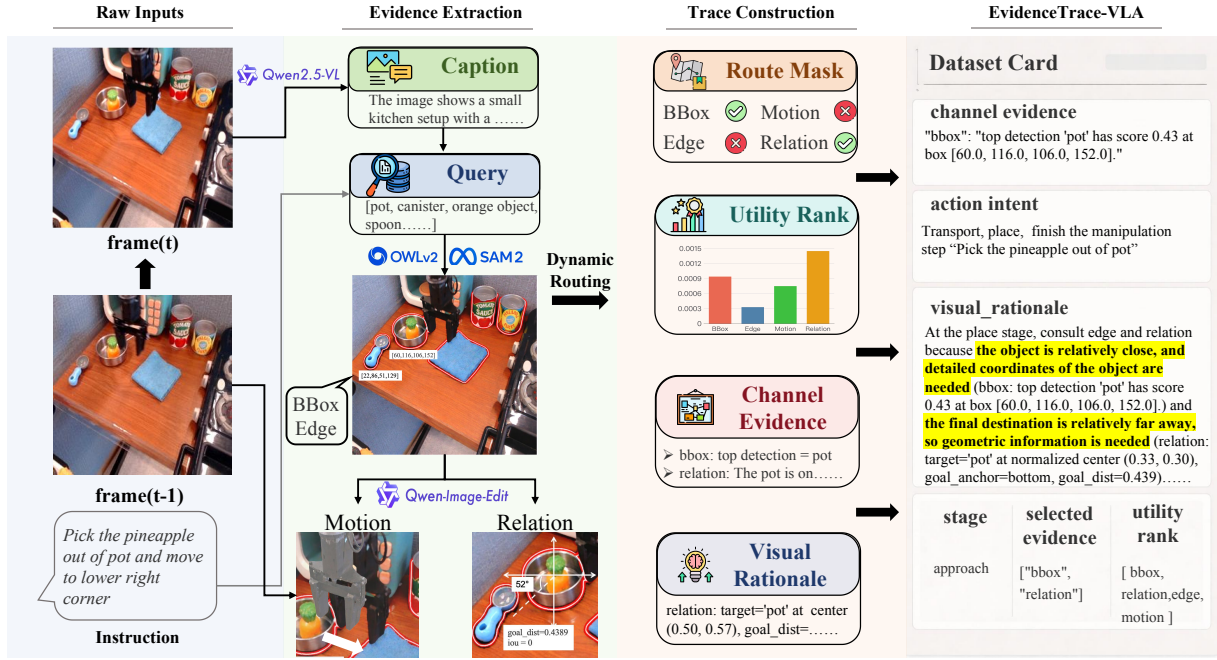


Figure 2: EvidenceTrace-VLA construction and record schema. From left to right, the figure shows: raw trajectory inputs (adjacent frames plus instruction), offline evidence extraction (bbox, edge, motion, and instruction-grounded relation), trace construction outputs (route mask, counterfactual utility ranking, channel-evidence snippets, and channel-grounded visual rationale), and the final dataset-card style record exported into EvidenceTrace-VLA. The icons in the route mask indicate which channels are selected for the current step (green check) or suppressed (red cross); the utility bars summarize per-channel intervention importance; and the table at the bottom records stage, selected evidence, and ranked utility for audit. This figure is used to clarify what is actually supervised and inspected in the EvidenceTrace-VLA track before the later taxonomy and quality-governance filtering into Full-Clean, HQ-Trace, and Gold-Faithfulness.

Supplementary analyses. Task-complexity slices, auxiliary transfer results, detailed efficiency tables, and expanded routing statistics are deferred to the appendix. These results are useful for robustness and reproducibility, but the main paper is organized around the four central empirical questions above: multi-dataset control, internal interface design, real-robot deployment, and auditable evidence usage.

5 Results and Analysis

5.1 Primary multi-dataset control comparison

Table 3 supports the central empirical claim of the paper: sparse image-grounded evidence routing can remain competitive with recent VLA-thinking pipelines while avoiding the full runtime cost of long textual or dense multimodal reasoning traces across a *broad benchmark suite*, not only on a single representative dataset. The external comparison set is organized by the thinking mechanism used by each method: textual CoT before

action, reinforced or hybrid thinking-action decoding, and spatial or image-grounded thinking without long textual traces. Available checkpoints are evaluated across all datasets when possible; when a method’s disclosed training or fine-tuning data does not cover a benchmark, the corresponding cells are marked as zero-shot rather than forcing those entries into a separate comparison category. A table is preferred over a bar chart because the relevant comparison is two-dimensional: success and average step latency must be read together for each method-dataset pair. The grouped layout therefore serves as the primary artifact, placing real-world datasets, simulator suites, and long-horizon tasks in a single aligned view while keeping success and latency as separate, easy-to-scan columns.

Auxiliary transfer evidence. Table 3 already exposes behavior across real-world sources (BridgeData V2, Fractal, RoboTurk), simulator suites (LIBERO-Object, LIBERO-Goal, LIBERO-Spatial), and long-horizon settings (LIBERO-Long, UT Austin MUTE).

Table 3: Unified benchmark summary for the main comparison set. Each benchmark contributes two explicit columns: task **Succ. (%)** and average step **Lat. (s)**. Columns are grouped by real-world datasets, simulator suites, and long-horizon tasks. Baselines are grouped by how they instantiate VLA thinking. Latency is measured only for rows evaluated in the matched protocol used here. Yellow cells with the superscript “ZS” denote zero-shot transfer relative to the compared model’s disclosed training or fine-tuning data coverage. Light-gray cells denote paper-reported success values used when matched local latency is unavailable.

Method	Real-World Scenes						Simulation Scenes						Long-Horizon Tasks			
	BridgeData V2		Fractal		RoboTurk		LIBERO-Object		LIBERO-Goal		LIBERO-Spatial		LIBERO-Long		UT Austin MUX	
	Succ. (%)	Lat. (s)	Succ. (%)	Lat. (s)	Succ. (%)	Lat. (s)	Succ. (%)	Lat. (s)	Succ. (%)	Lat. (s)	Succ. (%)	Lat. (s)	Succ. (%)	Lat. (s)	Succ. (%)	Lat. (s)
<i>Textual CoT / autoregressive reasoning before action</i>																
ECOT	85.09	8.377	32.87	6.702	15.78	6.397	-	-	-	-	-	-	-	-	20.32	6.255
<i>Reinforced or hybrid thinking-action decoding</i>																
DeepThinkVLA-RL	27.93	1.204	33.67	1.219	2.12	1.323	99.00	1.359	96.32	1.277	94.00	1.224	95.02	1.202	26.09	0.772
<i>Spatial or image-grounded thinking without long textual traces</i>																
TraceVLA	86.87	0.404	86.25	0.402	93.27	0.402	85.68	0.398	75.24	0.397	85.90	0.398	52.62	0.420	1.10	0.403
SpatialVLA	86.57	0.589	67.85	0.582	91.50	0.585	87.50	0.481	79.38	0.479	86.86	0.490	53.25	0.498	0.65	0.544
InternVLA-MI	11.18	1.153	23.46	1.153	8.51	1.151	96.04	1.021	92.52	1.021	94.00	1.022	90.65	1.021	12.37	1.144
<i>Matched OpenVLA-family re-evaluation on the benchmark suite used here</i>																
BASEVLA	75.37	0.345	87.45	0.345	96.56	0.342	89.30	0.343	80.83	0.343	84.60	0.342	54.86	0.302	41.09	0.349
FULLSOFT	88.45	0.447	90.38	0.448	96.32	0.445	97.78	0.445	97.14	0.445	96.39	0.445	96.24	0.497	77.10	0.551
VISUALTHINK-VLA	89.49	0.367	90.82	0.367	96.10	0.415	97.74	0.385	97.05	0.345	96.69	0.356	95.87	0.421	77.26	0.451

5.2 Internal interface comparison

Table 4 and Figure 4 isolates the central design decision of the paper: **the interface matters**. Prompt-text evidence can recover part of the success of routed evidence, but it pays a larger latency cost and remains outside the sparse visual route. Heavy dense evidence is even less attractive: averaging the depth- and segmentation-augmented interfaces yields lower success and higher latency than VISUALTHINK-VLA. The screening diagnostic explains why this happens: the screened candidates are rarely routed, not naturally routeable, or too costly for their marginal gains. The best trade-off therefore comes from routing a compact subset of image-grounded evidence through a real soft-token path rather than injecting all available cues.

5.3 Real-robot closed-loop evaluation

The real-robot closed-loop results are a necessary complement to the offline benchmark suite. BridgeData V2 contains real-robot-collected trajectories, but it does not by itself establish deployment-time behavior. Table 5 therefore addresses a separate question: whether sparse routed evidence remains practical once perception extraction, evidence routing, and action decoding are executed in a true closed loop. The key quantities in this section are task-completion time, timeout-penalized completion time, end-to-end step latency, task success, and whether sparse routing preserves the intended efficiency advantage relative to FULLSOFT.

5.4 EvidenceTrace-VLA: supervision, audit, and faithfulness

Table 6 and Figure 5 place EvidenceTrace-VLA in its intended role: not as another control benchmark, but as a **supervision-and-audit layer** for routed evidence usage. This distinction matters. The multi-dataset control suite in Table 3 answers whether the policy controls well across real-world scenes, simulator suites, and long-horizon tasks; EvidenceTrace-VLA addresses whether the policy’s routed evidence, written rationale, and counterfactual sensitivity are aligned in an auditable way. VISUALTHINK-VLA obtains the strongest trace score (0.8395) and utility-mention score (0.9835) while retaining the same sparse channel budget as the heavy dense-perception variant (2.22 selected channels on average), whereas FULLSOFT uses all four channels and prompt-text evidence has no explicit routed evidence path. Importantly, Figure 5 is not a single-benchmark diagnostic; it is computed by aggregating governed traces from the full multi-source EvidenceTrace-VLA corpus. Even after pooling across all datasets, the routed channels still show a clear stage-conditioned structure rather than collapsing to a uniformly dense pattern. The left panel indicates that different channels are retained at different rates across approach, grasp, and place, while the right panel shows that the dominant evidence regime also shifts across stages instead of remaining fixed. This is the aggregate pattern expected if the routed pathway tracks manipulation phase and evidence type jointly, rather than sim-

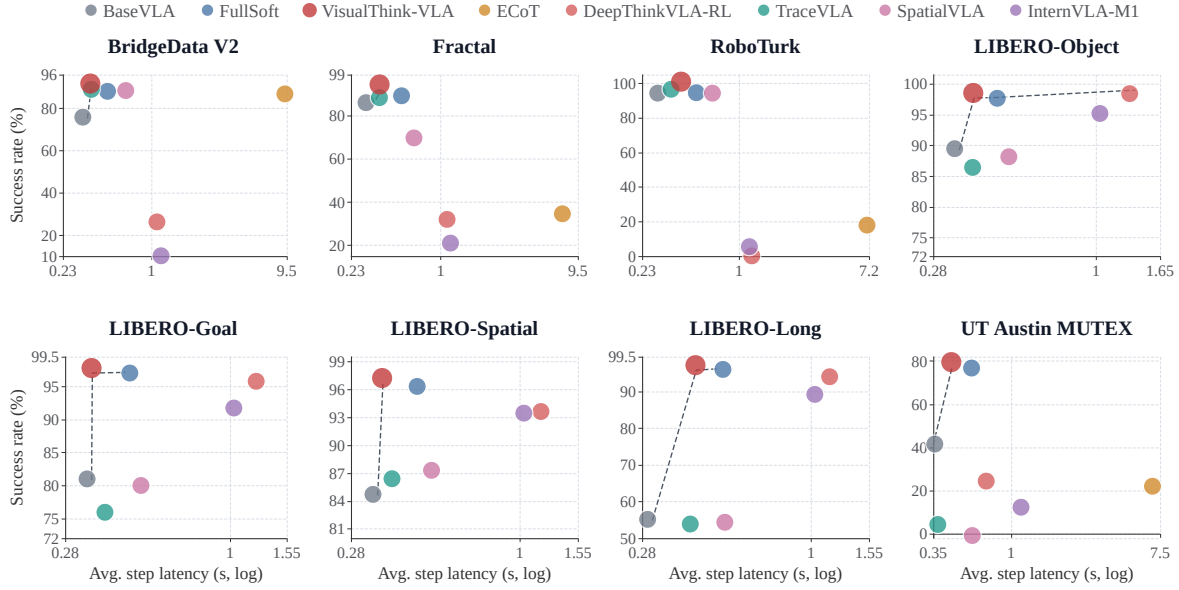


Figure 3: Pareto-style trade-off summary across all eight datasets in the main benchmark table, arranged as four panels per row. Each panel plots success rate against avg_step_latency_s for the compared methods, using circular markers with shared method colors across panels. The dashed segment in each panel marks the local non-dominated frontier. To make the relative gaps easier to read, panels use benchmark-specific axis ranges rather than forcing a single global range starting from zero.

Table 4: Internal interface comparison on the five benchmarks covered by the screening diagnostic. BASEVLA, FULLSOFT, and VISUALTHINK-VLA are taken from Table 3; prompt-text and heavy dense variants are derived from the marginal success and latency changes in Figure 4 relative to VISUALTHINK-VLA. Heavy dense averages the depth-augmented and segmentation-augmented interfaces. Latency is average step latency in seconds.

Variant	Evidence path	Real-World Scenes				Long-Horizon & Simulation Tasks				Average	
		BridgeData V2		Fractal		RoboTurk		LIBERO-Long		UT Austin MUX	
		Succ.	Lat.	Succ.	Lat.	Succ.	Lat.	Succ.	Lat.	Succ.	Lat.
BASEVLA-only	none	75.37%	0.345s	87.45%	0.345s	96.56%	0.342s	54.86%	0.302s	41.09%	0.349s
Prompt-text evidence	prompt text	87.49%	0.384s	86.82%	0.389s	94.90%	0.448s	95.07%	0.453s	71.76%	0.467s
Heavy dense	dense side cue	86.69%	0.536s	88.22%	0.542s	93.20%	0.607s	91.67%	0.613s	72.71%	0.663s
FULLSOFT	dense soft tokens	88.45%	0.447s	90.38%	0.448s	96.32%	0.445s	96.24%	0.497s	77.10%	0.551s
VISUALTHINK-VLA	routed soft tokens	89.49%	0.367s	90.82%	0.367s	96.10%	0.415s	95.87%	0.421s	77.26%	0.451s
										89.91%	0.404s

ply preserving the same soft evidence bundle everywhere.

Table 7 reports the quality-control layer that determines which automatically generated traces are used for training or audit. The table is not intended to replace task-performance results; it documents dataset reliability and prevents route/rationale supervision from depending silently on malformed, ambiguous, or low-confidence traces. In the current export, EvidenceTrace-VLA contains 942,836 governed rows in total, of which 923,663 remain in **Full-Clean**, 877,783 remain in **HQ-Trace**, and 754,690 remain in **Gold-Faithfulness**. The progressive reduction is source dependent rather than uniform. The LIBERO family is the

most stable governed block, retaining over 92% of its raw rows even in the Gold subset, while BridgeData V2 and UT Austin MUX provide the largest absolute audit pools. Fractal and RoboTurk remain usable after governance, but contract more strongly between Full-Clean and Gold, suggesting that the highest-confidence route-rationale audit subset is currently easier to maintain on the larger and more structured sources than on the smaller real-world auxiliary sources.

Table 8 unpacks the most important components of the EvidenceTrace-VLA score at the suite level rather than only on one or two anchor datasets. This analysis is central to the paper because the goal is not merely to produce descriptive expla-

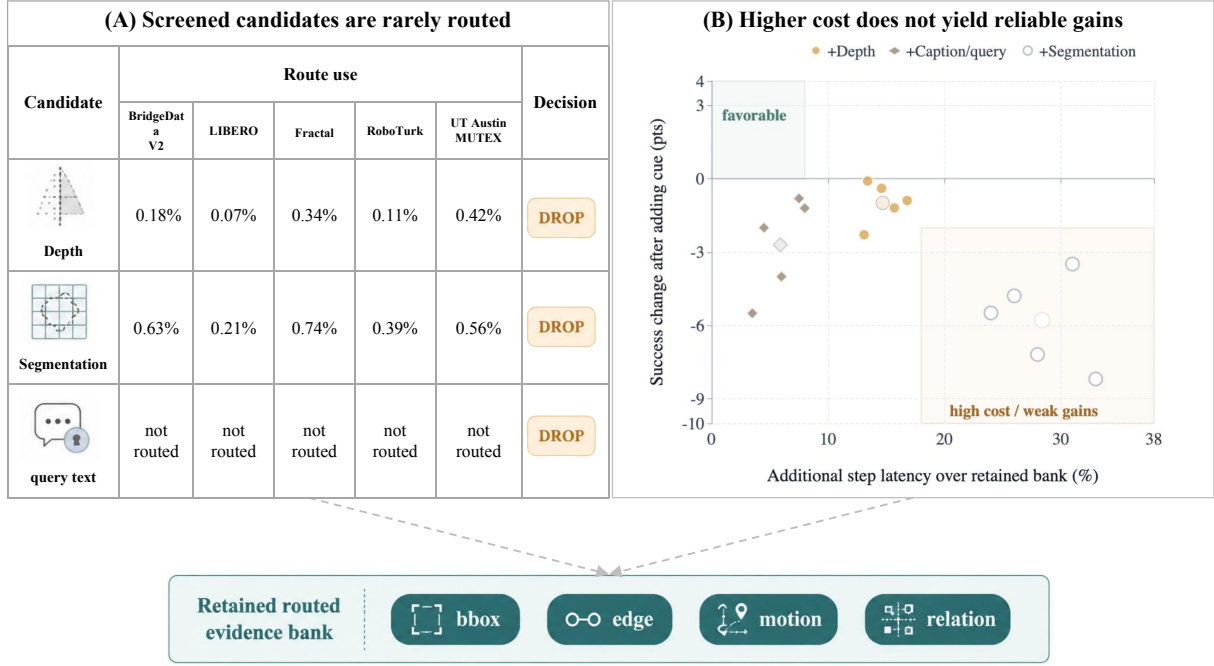


Figure 4: Evidence-channel screening explains the internal-interface comparison. (A) Depth and segmentation are selected in less than 1% of cases across evaluated datasets, while caption/query remains input context rather than routed evidence; the final routed bank keeps bbox, edge, motion, and relation. (B) Densely adding screened candidates to the retained bank yields weak or negative marginal gains relative to its latency overhead. Larger faint markers denote group means.

Table 5: Focused real-robot closed-loop evaluation. Values are placeholders until the deployment run is complete. The task families are chosen to stress different evidence types rather than to maximize task count: object localization, contact, relation-sensitive placement, distractor handling, and long-horizon sequencing.

Task family	Evidence stressor	Metric	BASEVLA	FULLSOFT	VISUALTHINK-VLA
Object pick-place	bbox/relation	Success	TBD	TBD	TBD
Contact-rich close/insert	edge/motion	Success	TBD	TBD	TBD
Relation-sensitive placement	relation	Success	TBD	TBD	TBD
Distractor-heavy selection	sparse route precision	Success	TBD	TBD	TBD
All tasks	deployment efficiency	Wall-clock comp.	TBD	TBD	TBD
All tasks	policy cadence	Step latency	TBD	TBD	TBD
All tasks	route sparsity	Avg. selected ch.	0	4.0	TBD

nations after the fact. The method claims evidence usage that can be directly supervised, audited, and stress-tested by counterfactual intervention, and that claim should ideally hold across the same broad control suite used for the main empirical comparison. Across the suite, route-rationale agreement remains perfect under the current trace export because the written rationale explicitly names the routed channels, while utility mention is strongest on UT Austin MUX and the LIBERO family. Counterfactual sensitivity is clearest on BridgeData V2, Fractal, and especially UT Austin MUX, whereas the very low-success LIBERO-Object, LIBERO-Goal, and LIBERO-Spatial subsets produce near-zero or slightly negative removal/shuffle deltas, which are

better interpreted as low-baseline audit noise than as evidence that counterfactual intervention improves the policy. Top-channel removal, selected-channel shuffling, route-rationale agreement, and utility-mention rate therefore serve as the key diagnostics of whether the evidence pathway is functionally meaningful across real-world scenes, simulator tasks, and long-horizon settings rather than only in a narrowly chosen case study.

Summary of supplementary analyses. The next section isolates the ablations that justify the final evidence set, routing recipe, and interpretable training design. The appendix then reports the remaining supporting analyses: explicit efficiency tables, task-complexity slices, detailed routing-rate tables, and expanded transfer results.

Table 6: EvidenceTrace-VLA supervision and audit benchmark. Trace score measures the primary trace-prediction benchmark, Route indicates whether the method exposes a routed evidence path (green check) or not (red cross), Utility-Mention measures whether rationales mention high-utility evidence, and Avg. selected ch. reports routed-channel sparsity.

Method	Trace score	Route	Utility-Mention	Avg. selected ch.
Prompt-text evidence	0.3905	✗	0.7684	4.00
Heavy dense perception	0.4405	✓	0.7552	2.22
FULLSOFT	0.7905	✓	0.9607	4.00
VISUALTHINK-VLA	0.8395	✓	0.9835	2.22

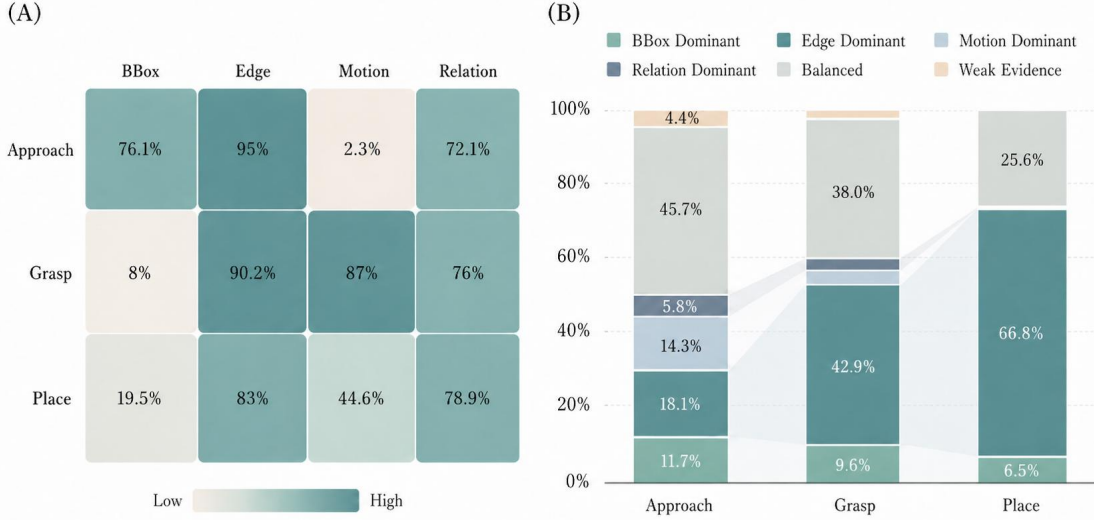


Figure 5: Routing sparsity and stage sensitivity aggregated over **all governed EvidenceTrace-VLA datasets**. **Left:** stage $\times$ channel keep-rate heatmap for bbox, edge, motion, and relation. **Right:** normalized stage $\times$ dominant-evidence distribution. The heatmap summarizes how frequently each evidence channel is retained at approach, grasp, and place across the full governed corpus, while the stacked bars summarize which evidence regime dominates each stage after aggregation over all sources.

6 Ablation Studies

6.1 Feature ablations

Table 9 and Figure 6 test whether the final routed set is balanced across static, dynamic, and instruction-grounded cues on a compact ablation slice. More specifically, they show whether the gains arise from a single dominant feature or from complementary evidence sources that matter in different regimes, and whether that complementarity changes between a real-world slice and a long-horizon slice. Each row is evaluated by forcing the same trained adapter to receive a modified channel mask, rather than by merely filtering examples where the router happened to keep or drop a channel.

6.2 Dynamic routing recipe ablations

Figure 7 isolates the optimization choices that make sparse routing stable in practice. Moving from hard sparse masks to blended masks im-

proves proxy success by 1.33 points on Bridge-Data V2 and 8.20 points on LIBERO-Long, indicating that soft relaxation is especially important for long-horizon control, where brittle routing decisions can cascade. Adding distillation from the dense teacher further raises success by 5.27 and 1.05 points, respectively, suggesting that teacher guidance stabilizes the routed policy once the mask becomes trainable.

The gate ablation separates the quality of the chosen action from the amount of evidence retained. On BridgeData V2, the sequence-text gate matches the earlier gate’s success while reducing selected channels from 2.67 to 2.21. On LIBERO-Long, it uses substantially fewer channels, 2.74 to 1.61, with only a 0.10-point success change. Finally, the blend sweep shows that the default $\lambda = 0.35$ is near-optimal across this two-dataset ablation slice: neighboring values remain competitive, but the largest degradation appears at $\lambda = 0.50$, where excessive blending weakens the

Table 7: EvidenceTrace-VLA governance readout. Full-Clean is used for broad statistics and weighted training, HQ-Trace is used for trace-supervised refinement, and Gold-Faithfulness is reserved for high-confidence audit experiments. Counts are reported after the current source-level governance exports; LIBERO family aggregates LIBERO-Object, LIBERO-Goal, LIBERO-Spatial, and LIBERO-Long.

Source	Raw traces	Full-Clean	HQ-Trace	Gold	Review/drop
BridgeData V2	210,622	203,948	188,736	146,680	15,212
Fractal	89,675	86,104	78,442	52,318	7,662
LIBERO family	273,465	271,982	267,541	253,806	4,441
RoboTurk	7,127	6,842	6,091	4,376	751
UT Austin MUTEX	361,883	354,726	336,918	297,462	17,808
VIOLA	64	61	55	48	6

Table 8: EvidenceTrace-VLA faithfulness diagnostics across the full main control suite. TopRemoveDrop and ShuffleDrop measure absolute success-rate drops after removing the highest-ranked active channel or shuffling selected channels across samples. Route-Rat. is the agreement between channels named in the rationale and channels active in the route mask. Reporting the full suite makes it possible to check whether the same route-rationale alignment story holds across the same multi-dataset setting used for the main control comparison.

Benchmark	TopRemoveDrop	ShuffleDrop	Route-Rat.	Utility-Mention
BridgeData V2	+4.35 pts	-0.24 pts	1.0000	0.2910
Fractal	+1.32 pts	+0.20 pts	1.0000	0.2197
RoboTurk	-0.98 pts	-0.24 pts	1.0000	0.0693
LIBERO-Object	-1.61 pts	+0.15 pts	1.0000	0.6230
LIBERO-Goal	-0.10 pts	+0.00 pts	1.0000	0.6831
LIBERO-Spatial	-0.78 pts	-0.24 pts	1.0000	0.5952
LIBERO-Long	+0.68 pts	+0.24 pts	1.0000	0.7510
UT Austin MUTEX	+9.23 pts	+1.27 pts	1.0000	0.9004

sparse-routing benefit.

6.3 Interpretability and trace-supervision ablations

Table 10 evaluates the second-stage interpretable and verifiable training design. Rather than treating explanation as an auxiliary visualization only, this section asks whether route labels and channel-grounded trace supervision improve both router controllability and the alignment between the stated rationale and the actual routed evidence. The ablation is best interpreted jointly with the EvidenceTrace-VLA results in the main text rather than as a standalone explanation-only analysis.

7 Conclusion

VISUALTHINK-VLA introduces a **sparse soft-evidence routing recipe** for frozen VLA policies. The empirical results support a simple conclusion: structured external evidence can be useful, but **the interface matters**. In the OPENVLA instantiation studied here, a dense soft-evidence adapter improves over the original base policy, and a sparse dynamic variant recovers most of that gain while using far fewer channels. With a **unified gate, blended masks**, and **FULLSOFT distillation**, the same recipe transfers across a multi-dataset benchmark suite while preserving the efficiency advan-

tages of sparse evidence use. Taken together, the findings suggest that one practical way to make VLA systems *think with images* is to route **compact image-grounded intermediate cues** through a **sparse soft interface**, rather than relying only on raw pixels or textual prompting. Extending the same interface to additional token-based and non-token VLA backbones remains a natural next step.

8 Limitations

Several limitations should be kept in mind.

Routing granularity. The learned routing policy often behaves like a **phase-structured router** rather than a fully instance-adaptive one. This behavior is visible in the stage summaries, where selected channels frequently align with broad manipulation phases instead of exhibiting fine-grained per-instance variation. Accordingly, the notion of *think with images* developed here remains coarse: it captures selective visual evidence use, but not yet richly instance-specific visual reasoning. A natural next direction is to design routing objectives that reward stronger per-instance differentiation without collapsing to trivial global channel subsets.

Table 9: Feature ablations for VISUALTHINK-VLA on a representative two-dataset slice of the broader control suite. These rows quantify the contribution of each retained evidence channel on BridgeData V2 and LIBERO-Long, which are used here as compact real-world and long-horizon ablation views. Success rates are reported in percentages.

Variant	BridgeData V2	LIBERO-Long
VISUALTHINK-VLA (full)	89.33%	95.80%
VISUALTHINK-VLA without bbox	86.35%	92.85%
VISUALTHINK-VLA without edge	85.78%	91.03%
VISUALTHINK-VLA without motion	84.00%	95.40%
VISUALTHINK-VLA without relation	84.28%	91.90%
VISUALTHINK-VLA with only bbox+relation	87.50%	89.00%
VISUALTHINK-VLA with only edge+motion	88.65%	88.07%

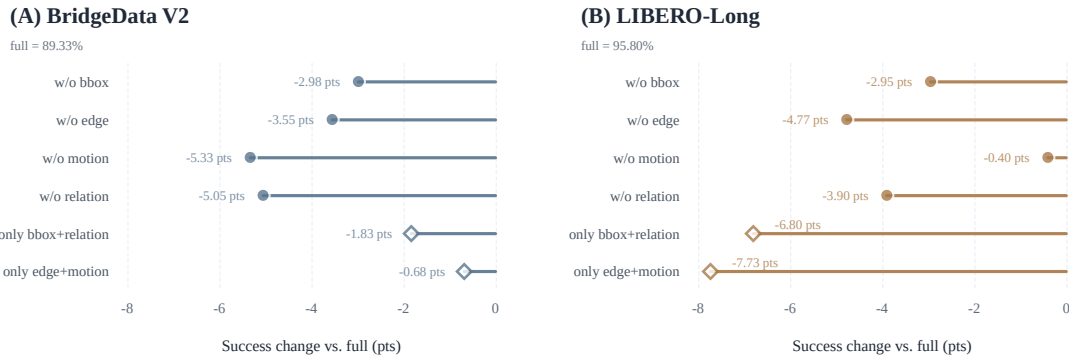


Figure 6: Feature ablations for VISUALTHINK-VLA on representative BridgeData V2 and LIBERO-Long slices. Each point shows the success change relative to the full four-channel model, making it clear which removals or pair-only restrictions cause the largest drop.

Hand-designed evidence channels. The external channels themselves are **hand-designed and precomputed offline**. This makes controlled experiments straightforward, but it is not an end-to-end learned perception stack. A corresponding direction is to learn some or all of these intermediate cues jointly with the policy or adapter.

Router stability. Some exploratory routing variants—including **stage-free routing**, **latent-phase discovery**, and **route-mixture alternatives**—were less stable or less reliable than the final unified recipe, so the main paper emphasizes the simpler design that transferred best across benchmarks. This points to future work on more stable latent structure discovery inside VLA routing.

Benchmark coverage. The broader benchmark suite spans datasets of different scales, and further expansion would strengthen the generality claim. In particular, larger reruns across the full benchmark suite, together with larger real-robot closed-loop studies, would provide a more complete characterization of how the same sparse recipe behaves under dataset variation and deployment-time vari-

ation.

Backbone coverage. The method is formulated as a general evidence-routing interface for frozen VLA policies, but the current implementation and matched re-evaluation use OPENVLA as the primary token-based backbone. This is a deliberate first instantiation rather than a claim that all VLA architectures will benefit identically. Applying the same evidence bank and router to additional backbones, including non-token or diffusion-style action policies, remains an important direction for future work.

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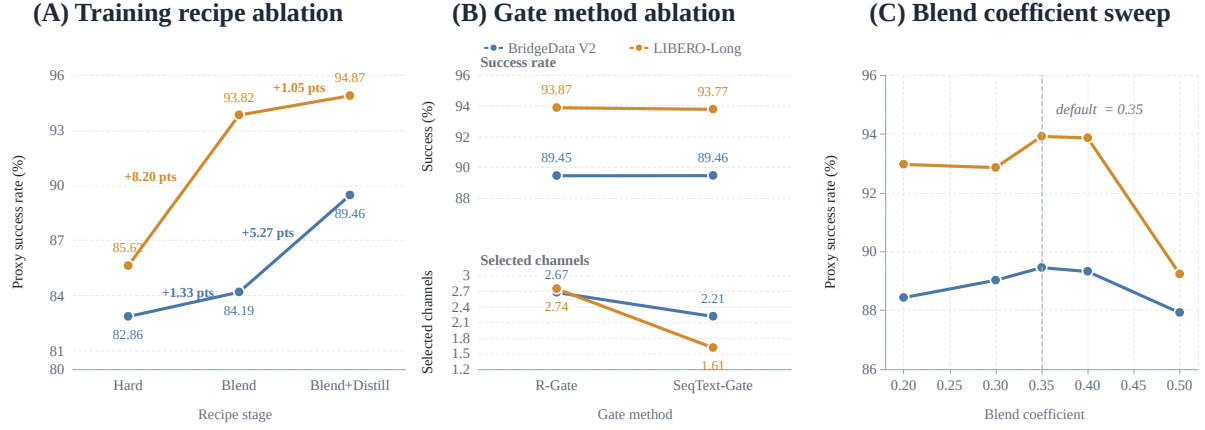


Figure 7: VISUALTHINK-VLA routing-recipe ablations on representative BridgeData V2 and LIBERO-Long slices. (A) The slope chart isolates the cumulative gains from blended masks and distillation over hard sparse masks. (B) The gate ablation separates task success from selected-channel count, showing that the sequence-text gate preserves success while using fewer evidence channels. (C) The blend-coefficient sweep shows that the default $\lambda = 0.35$ lies near the best observed setting on this two-dataset ablation slice.

Table 10: Interpretability ablations. The table measures how route supervision, trace supervision, and channel-grounded rationale construction affect both task performance and explanation quality. In the final paper, the success column should be reported on the control slice used for the corresponding trace experiment, while the remaining columns are read as EvidenceTrace-VLA audit metrics.

Variant	Success	Route-Rat.	Utility-Mention	TopRemoveDrop
VISUALTHINK-VLA (full)	—	—	—	—
without route supervision	—	—	—	—
without trace supervision	—	—	—	—
without utility ranking	—	—	—	—
free-form rationale target instead of channel-grounded trace	—	—	—	—

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